

The property of the bisector of an angle

Every point of a bisector is the same distance from the two arms — and only those points are.

LESSON 90–91

2 × 40 MIN

MATEMATIKA 7, §42

S8 5.3

LESSON CLOCK

10'

25'

20'

25'

5'

What distance means here	10 min
The property	25 min
The converse	20 min
The construction and its use	25 min
Homework	5 min

LEARNING OBJECTIVES

- State the bisector property and its converse.
- Prove the property using congruent right-angled triangles.
- Construct the bisector of an angle with compasses.
- Use the property to find the point equidistant from two lines.

TEXTBOOK

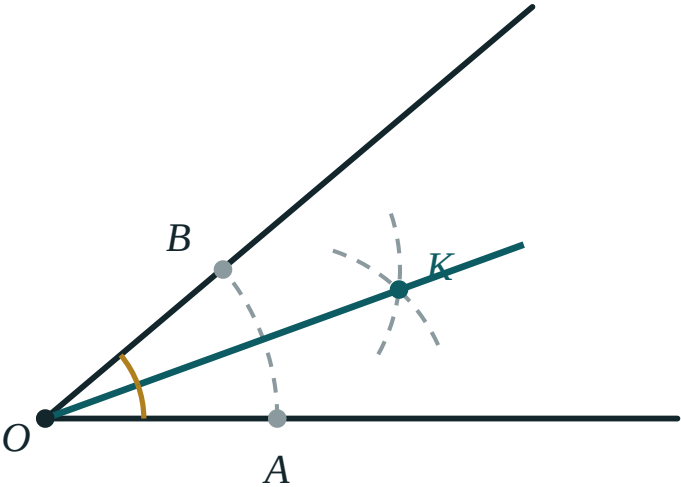
National	pp. 262–268
Cambridge	Stage 8, pp. 52–56
Workbook	Exercise 5.3

01

Explanation

What distance means here

The distance from a point to a line is the length of the perpendicular dropped from the point to the line — not any slanting segment.



Dropping perpendiculars to both arms

SEGMENT FROM M TO THE LINE	IS IT THE DISTANCE?
perpendicular	yes — and it is the shortest
slanting	no — it is longer

MEASURE TO THE ARM, NOT TO THE VERTEX

The distance from a point on the bisector to the vertex is not what the theorem is about. Every statement below concerns the perpendiculars to the two arms.

The property

Let M lie on the bisector of $\angle AOB$, and drop perpendiculars MP and MQ to the arms.

STATEMENT	REASON
$\angle MOP = \angle MOQ$	OM is the bisector
$\angle MPO = \angle MQO = 90^\circ$	construction
$OM = OM$	common hypotenuse
$\triangle MPO \equiv \triangle MQO$	hypotenuse and an acute angle
$MP = MQ$	corresponding sides of congruent triangles

M on the bisector $\implies MP = MQ$

THE RIGHT-ANGLE CONGRUENCE TEST EARNS ITS KEEP

The proof needs exactly the test met last lesson: a common hypotenuse and one equal acute angle. That is why the two topics stand next to each other in the plan.

The converse

The statement also runs backwards: a point inside the angle that is equidistant from both arms lies on the bisector.

DIRECTION	GIVEN	CONCLUSION
property	M	$MP = MQ$
converse	$MP = MQ$	M

Taken together the two say: the bisector is **exactly** the set of points inside the angle equidistant from the arms — a locus, not merely a line with a nice property.

WHERE THE THREE BISECTORS MEET

Each bisector of a triangle is the locus of points equidistant from two sides, so the point common to two of them is equidistant from all three — and therefore lies on the third. That point is the incentre, the centre of the inscribed circle.

The construction and its use

STEP	WHAT TO DO
1	with centre O , cut both arms at A and B
2	with the same radius and centres A and B , draw two arcs
3	call their intersection M
4	draw OM — it bisects the angle

Why it works: $OA = OB$ and $AM = BM$ by construction, and OM is common, so $\triangle OAM \equiv \triangle OBM$ by the third criterion and the two angles at O are equal.

PROBLEM	THE POINT WANTED
a well equidistant from two straight roads	on the bisector of the angle between them
a lamp lighting two walls equally	on the bisector of the corner
the centre of a circle touching both arms	on the bisector

02 Worked examples

EXAMPLE 1 M lies on the bisector of $\angle AOB$ and its distance to one arm is 3.5 cm . Find its distance to the other.

- ① The bisector property applies.
- ② Both perpendiculars are equal.
- ③ So the distance is 3.5 cm .

ANSWER

3.5 cm

EXAMPLE 2 A point inside an angle is 4 cm from each arm. What can be said about it?

- ① It is equidistant from both arms.
- ② By the converse, it lies on the bisector.

ANSWER

It lies on the bisector

EXAMPLE 3 Prove that the compass construction really bisects the angle.

- ① $OA = OB$
Same radius from O .

- ② $AM = BM$
Same radius from A and B .

- ③ $OM = OM$
Common side.

- ④ $\triangle OAM \cong \triangle OBM$ by the third criterion, so $\angle AOM = \angle BOM$.

ANSWER

Proved by SSS

03 Interactive model

Ask where to dig a well equidistant from two roads before any theory; the class reaches for the bisector on its own, and the proof then explains a choice they have already made.

Distance, bisector, locus

The distance from a point to a line is measured:

any way

along the perpendicular

to the nearest end

to the vertex

A point on a bisector is equidistant from:

the two arms

the vertex and an arm

two vertices

nothing

The proof of the property uses:

the third criterion

hypotenuse and an acute angle

the angle sum

the axiom of parallels

A point equidistant from both arms lies:

at the vertex

on the bisector

anywhere inside

outside

The bisector is the set of such points — that makes it a:

locus

median

transversal

tangent

The three bisectors of a triangle meet at the:

centroid

incentre

circumcentre

orthocentre

The compass construction is proved by:

SSS

SAS

ASA

measuring

A well equidistant from two roads should be:

at the crossing

on the bisector

on either road

anywhere

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Bisector	Bissektrisa	Биссектриса
Arm of an angle	Burchak tomoni	Сторона угла
Equidistant	Teng uzoqlikda	Равноудалённый
Distance to a line	Nuqtadan chiziqqa masofa	Расстояние до прямой
Converse	Teskari teorema	Обратная теорема
Locus	Geometrik o'rin	Геометрическое место точек
Compasses	Sirkul	Циркуль
Incentre	Ichki markaz	Центр вписанной окружности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A bisector divides an angle into:

two equal parts

two unequal parts

three parts

a right angle

Distance to a line is measured along the:

bisector

perpendicular

arm

diagonal

On a bisector, the two perpendicular distances are:

equal

in ratio 2 : 1

unrelated

zero

The converse concludes that the point:

is the vertex

lies on the bisector

is outside

is the midpoint

The proof uses which congruence test?

two legs

hypotenuse and an acute angle

three angles

two angles only

The incentre is equidistant from:

the vertices

the three sides

two sides

the midpoints

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A bisector divides an angle into

ANSWER two equal angles

2. Distance from a point to a line is measured along the

ANSWER perpendicular

3. M on the bisector, one distance 3.5 cm : the other

ANSWER 3.5 cm

4. A point 4 cm from each arm lies on the

ANSWER bisector

5. The bisector of 80° makes two angles of

ANSWER 40°

6. The three bisectors of a triangle meet at the

ANSWER incentre

7. The incentre is the centre of the

ANSWER inscribed circle

Medium · the standard question

7 PROBLEMS

1. The bisector of $\angle AOB = 116^\circ$: each half

ANSWER 58°

2. Two bisectors of adjacent supplementary angles meet at

ANSWER 90°

3. Name the congruence test used in the proof

ANSWER Hypotenuse and an acute angle

4. Name the congruence test used for the construction

ANSWER The third criterion (SSS)

5. Where should a well equidistant from two roads be dug?

ANSWER On the bisector of the angle between them

6. Is the vertex equidistant from both arms?

ANSWER Yes — both distances are zero

7. A point is 3 cm from one arm and 5 cm from the other: is it on the bisector?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. Prove that the bisector property and its converse together define a locus

ANSWER Every point on it qualifies and every point that qualifies is on it

2. Why do the three bisectors of a triangle meet at one point?

ANSWER Two give a point equidistant from all three sides, so the third passes through it

3. In a triangle with $\angle A = 60^\circ$, the bisectors of $\angle B$ and $\angle C$ meet at I : find $\angle BIC$

ANSWER 120°

4. The bisector of the apex of an isosceles triangle is also its

ANSWER median and altitude

5. A circle of radius 2 cm touches both arms: where is its centre?

ANSWER On the bisector, 2 cm from each arm

6. Explain why the compass construction cannot fail for an angle under 180°

ANSWER The two equal arcs always meet inside the angle

7. The bisectors of two vertical angles form

ANSWER one straight line

07 Homework

Homework — 5 tasks

Leave your construction arcs on the page — they are the working.

- Construct the bisector of an angle of 70° with compasses and check it with a protractor.
- Write the proof of the bisector property in statement-and-reason form.
- A point inside an angle is 2.5 cm from each arm. Where does it lie, and why?
- Construct the three bisectors of a triangle and mark the incentre.
- Explain why the bisector is called the locus of points equidistant from the arms.